📦 Order Statistics: The i-th Smallest Value

Order statistics deals with the probability distributions of the values when a random sample is sorted.

1. Definition

Given a random sample of n independent and identically distributed (i.i.d.) random variables, $X_1, X_2, \ldots, X_n$, we define the **order statistics** as the sample values sorted in non-decreasing order:

$$X_{(1)} \leq X_{(2)} \leq \ldots \leq X_{(n)}$$

- $X_{(1)}$ is the **minimum** (the 1st smallest value).
- $X_{(n)}$ is the **maximum** (the n-th smallest value).
- $X_{(i)}$ is the **i-th order statistic** (the i-th smallest value).

2. Finding the CDF of the i-th Order Statistic, $X_{(i)}$

To find the distribution of $X_{(i)}$, we first look at its **Cumulative Distribution Function (CDF)**, $F_{X_{(i)}}(x) = P(X_{(i)} \leq x)$.

Intuition: The i-th smallest value, $X_{(i)}$, is less than or equal to some value x if and only if **at least i** of the original n variables ($X_1, \ldots, X_n$) are less than or equal to x.

This converts the problem into a **Binomial distribution** context:

- **Number of trials:** n (the sample size).
- **Success:** The event that an individual variable X_j is $\leq x$.
- **Probability of success:** $p = P(X_j \leq x) = F(x)$, where $F(x)$ is the CDF of the original distribution.

Thus, the CDF of $X_{(i)}$ is the probability of getting k successes (where $k \geq i$) in n Bernoulli trials:

$$F_{X_{(i)}}(x) = \sum_{k=i}^{n} P(\text{exactly } k \text{ values are } \leq x)$$

$$F_{X_{(i)}}(x) = \sum_{k=i}^{n} \binom{n}{k} [F(x)]^k [1 - F(x)]^{n-k}$$

3. Finding the PDF of the i-th Order Statistic, $X_{(i)}$

The **Probability Density Function (PDF)**, $f_{X_{(i)}}(x)$, is found by differentiating the CDF above, which involves a lot of messy algebra. A cleaner, more intuitive approach is to consider the event in a small neighborhood, dx.

Intuition: For the i-th smallest value $X_{(i)}$ to fall precisely at x (in the interval $[x, x + dx]$), three things must happen simultaneously:

1. **Exactly one** variable must fall in the small interval $[x, x + dx]$.
   - Probability: $f(x) \cdot dx$
2. **Exactly $i - 1$** variables must be less than x.
   - Probability: $[F(x)]^{i-1}$
3. **Exactly $n - i$** variables must be greater than x.
   - Probability: $[1 - F(x)]^{n-i}$

We must also count the number of ways (arrangements) these three groups can be selected from the n variables. This is a **multinomial coefficient**: $\frac{n!}{(i-1)! \cdot 1! \cdot (n-i)!}$.

Combining these elements gives the general PDF formula for the i-th order statistic:

$$f_{X_{(i)}}(x) = \frac{n!}{(i-1)!(n-i)!} [F(x)]^{i-1} [1 - F(x)]^{n-i} f(x)$$

4. Special Cases

This general formula simplifies nicely for the minimum and maximum values:

A. Minimum $X_{(1)}$

Substitute $i = 1$ into the PDF formula ($i - 1 = 0, n - i = n - 1$):

$$f_{X_{(1)}}(x) = n[1 - F(x)]^{n-1} f(x)$$

B. Maximum $X_{(n)}$

Substitute $i = n$ into the PDF formula ($i - 1 = n - 1, n - i = 0$):

$$f_{X_{(n)}}(x) = n[F(x)]^{n-1} f(x)$$

This concludes the detailed breakdown of the three key concepts from your notes: Independence/Covariance, Convolution, and Order Statistics.

Do you have any specific examples you'd like to work through for these concepts, or would you like to move on to a new topic?